

Amit Gohel

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Professional Summary

AI/ML Engineer with 2+ years in production LLM systems – agentic RAG pipelines (LangGraph + Milvus, 95% MRR, 50K+ queries/month), LLM inference optimization (vLLM, 8s under 500+ concurrent users), and audio/speech ML research (CNN-BiLSTM, 1D CNN+GRU, IIT Delhi, Samsung Solve for Tomorrow Top 10). HuggingFace Transformers across 1,000+ empirically evaluated model configurations. Microsoft Azure AI Engineer Associate certified.

Work Experience

AI Engineer, Vivansh Infotech *Ahmedabad, India* Aug 2025 – Present

- Upgraded permission-aware RAG chatbot to fully **agentic architecture using LangChain/LangGraph** for multi-tenant EHS platform serving **500+ concurrent users, 1,000+ queries/day**; integrated Milvus, PostgreSQL, and 10+ platform APIs for autonomous data retrieval and workflow triggers
- Enforced **250+ granular permissions per query** via 2-stage retrieval: dedicated Milvus permission index cross-referenced with PostgreSQL, with GPT-4o handling access control, query rewriting, and conversation resolution across 10,000+ chunks; **80% MRR**
- Built **AI-powered form builder** converting PDF forms to digital replicas in ~20 seconds (**200+ forms in month 1**), bulk processing 100+ PDFs in 5–10 minutes; replaced 1+ hour of manual work per form via Pydantic-validated intermediate JSON and multi-model pipeline (GPT-4o, Qwen3-Coder-480B, GPT-4o-mini)
- Created **AI video course generation pipeline** reducing production from ~1 week to ~10 minutes (**99% reduction**) via RAG-based video selection (Milvus + Cohere), Gemini 2.5 Pro for visual-audio alignment, ElevenLabs narration, and serverless FFmpeg rendering on AWS Lambda

AI/ML Developer, Silver Touch Technologies Ltd *Ahmedabad, India* Jun 2024 – Jul 2025

- Rebuilt production RAG chatbot infrastructure serving **50,000+ queries/month** across 10+ chatbot instances; migrated from in-code Llama-2-7b to VLLM-served API and FAISS to Milvus, improving **MRR by 111% (45% to 95%)**
- Led vector DB evaluation (Milvus vs. Qdrant vs. FAISS) on production-scale data; tested 6+ embedding models (BGE-M3, BGE-small/base/large, Stella, SPLADE) and built custom retrieval with BGE-M3 + reranker
- Reduced response latency by **50% (16s to 8s)** through query preprocessing with intent identification and retrieval pipeline optimization
- Built **on-premise OCR and document pipeline** for sensitive government documents (200+ page PDFs); merged DocTR + DocLink via Llama-3.1-Nemotron-70b, achieving **~90% entity extraction accuracy** on Aadhaar, PAN, and GST numbers

Research Intern, Indian Institute of Technology (IIT) Delhi *New Delhi, India* Jan 2024 – May 2024

- Built end-to-end **air writing recognition system**: ESP32 + MPU6050 wearable capturing 6-axis IMU data at 60Hz, classifying 62 alphanumeric characters via CNN-BiLSTM with **75% user-dependent accuracy**
- Collected 12,400 labeled samples from 20 participants; evaluated **1,000+ model configurations** across 5 architecture families (Dense, CNN, LSTM, BiLSTM, hybrids) using TensorFlow/Keras with leave-one-subject-out cross-validation; collaborated with IISc Bangalore and Howard University researchers

Technical Skills

- LLMs and Techniques**: GPT-4o, Gemini 2.5 Pro, Llama, Qwen3-Coder; SFT, LoRA/QLoRA, LLM Fine-tuning, RAG, Agentic AI, NLP, CNN, LSTM, BiLSTM, GRU, TinyML
- Vector DBs and Retrieval**: Qdrant, Milvus, FAISS; BGE-M3, Cohere Embed/Rerank, OpenAI Embeddings, SPLADE, Stella
- Languages and Frameworks**: Python, C++, SQL; PyTorch, TensorFlow/Keras, Hugging Face Transformers/TRL, LangChain, LangGraph, FastAPI, Pydantic
- Infrastructure and Tools**: AWS (EC2, Lambda, S3, SAM), Docker, Git, PostgreSQL, vLLM, DocTR, OpenAI/Gemini/Anthropic APIs, Together AI, ElevenLabs, FFmpeg

Projects

Voicer – Sign Language to Speech Device (GitHub) 2023

- Wearable device translating Indian Sign Language to speech using 10 IMU sensors + NodeMCU with on-device TinyML inference in **2ms, 94.1% accuracy** across 12 gesture classes; **Samsung Solve for Tomorrow Top 10 among 70,000+ participants**, featured on CNN News18

WeepScope – Infant Cry Classifier (GitHub) 2023

- Standalone device classifying infant cries into 5 categories using hybrid 1D CNN + GRU on Arduino Nano 33 BLE Sense (1MB flash); **86% test accuracy** with 5-cycle majority voting; co-authored research paper with Dr. Kirankumar Trivedi

Education

B.E. Computer Engineering, Vishwakarma Government Engineering College, Ahmedabad Jun 2020 – Jun 2024

Certifications and Achievements

- Microsoft Azure AI Engineer Associate** – Certified in designing and implementing Azure AI solutions
- Smart India Hackathon 2023** – Top 5 among 44,000+ teams nationwide (Hardware Edition, MIT Karnataka)
- Competitive Programming**: 370+ problems solved in Python